QBUS5001 GROUP ASSIGNMENT

ASSIGNMENT COVER PAGE

Group Number: 78

Student ID	Student name	Contribution Percentage (0% - 100%)
550297954	Ziyan Wang	100
540793633	Xueru Teng	100
550209083	Junyu Shen	100
550051459	Yezhou Liang	100

Declaration

By signing below, each member confirms their contribution to the group effort as indicated by the percentage provided.

Signature and Date:

Xueru Teng 2025.5.24

Ziyan Wang 2025.5.24

24/05/2025

Junyu Shen 2025.5.24

Task 1

1. Data simulation and method description

The goal of this task is to verify the central limit theorem through the sample mean of the exponential distribution Exp(λ=0.2). Since Excel does not directly support the generation of random numbers for exponential distribution, Excel Analysis ToolPak is used to generate random numbers. The specific method is as follows: First create a new sheet in Excel and name it Task 1 (Random sample). Use "Data" > "Data Analysis" > "Random Number Generation", and then fill in the conditional information according to the task requirements to generate random numbers (see Figure 1). Finally, click OK to obtain a set of 100 x 1000 random number samples (see Figure 2). Among them, using the uniform distribution of U(0.01,0.99) for simulation can effectively avoid generating extremely small or extremely large values, making the final conclusion more accurate.

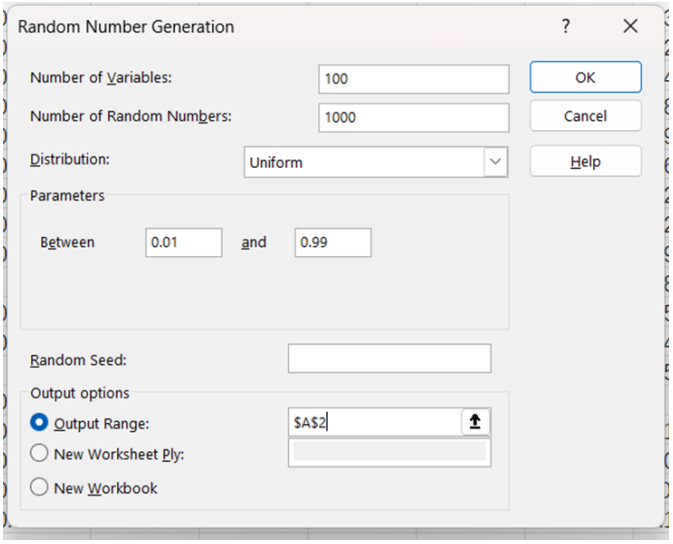


Figure 1. Excel Analysis ToolPak generates random numbers

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Simulate a random sample from the uniform distribution [U(0.01,0.99)]														
2	0.133251	0.380143	0.617315	0.625958	0.709192	0.673213	0.966402	0.48201	0.347274	0.352299	0.852871	0.339947	0.61543	0.919596	0.1826
3	0.208261	0.292213	0.334085	0.584505	0.323856	0.167436	0.38122	0.408107	0.087103	0.888282	0.148804	0.209427	0.448124	0.458682	0.61241
4	0.076576	0.556482	0.440827	0.944061	0.791141	0.032969	0.827868	0.142373	0.909936	0.35846	0.655806	0.475281	0.631342	0.482818	0.030128
5	0.785398	0.014905	0.911282	0.034345	0.118417	0.306778	0.907753	0.393542	0.860378	0.686402	0.100771	0.894294	0.051991	0.549633	0.818088
6	0.90527	0.383792	0.882749	0.815307	0.560848	0.509915	0.348859	0.733867	0.542694	0.626437	0.02947	0.952525	0.131756	0.184394	0.379485
7	0.661818	0.841237	0.805766	0.630385	0.772598	0.775499	0.174405	0.195939	0.962664	0.400241	0.349667	0.678447	0.060246	0.563091	0.421835
8	0.931918	0.181194	0.51479	0.057046	0.248697	0.406193	0.783066	0.214302	0.215469	0.975883	0.146501	0.278904	0.459639	0.501122	0.983301
9	0.565005	0.07185	0.845813	0.136661	0.407629	0.629098	0.176469	0.264847	0.320058	0.347244	0.636187	0.221092	0.811299	0.375806	0.057554
10	0.880476	0.506296	0.255994	0.877396	0.875123	0.186189	0.940412	0.565125	0.710299	0.442801	0.733029	0.933563	0.965116	0.660921	0.563151
11	0.221062	0.584266	0.931948	0.692952	0.800352	0.329	0.773615	0.228838	0.464215	0.560669	0.695913	0.809415	0.882211	0.860916	0.432512
12	0.153828	0.61232	0.160797	0.850628	0.824578	0.014725	0.651709	0.337674	0.481263	0.759259	0.409513	0.505159	0.111598	0.223873	0.886548
13	0.318473	0.041134	0.071401	0.871295	0.833192	0.812555	0.520622	0.323288	0.248637	0.900664	0.906197	0.457575	0.951419	0.14016	0.83047
14	0.574008	0.947919	0.739789	0.715025	0.871803	0.35825	0.925518	0.062758	0.987219	0.155623	0.583847	0.592551	0.135016	0.785937	0.676264
15	0.184484	0.801489	0.945257	0.595093	0.356695	0.724984	0.070863	0.980908	0.017537	0.307795	0.60568	0.9085	0.433619	0.738831	0.187475
16	0.934431	0.879489	0.972863	0.533452	0.020677	0.748372	0.797541	0.921989	0.876828	0.025463	0.157148	0.147577	0.554657	0.944689	0.912329
17	0.973969	0.354063	0.387411	0.301006	0.744664	0.634362	0.311773	0.584535	0.507163	0.099126	0.946214	0.057973	0.607565	0.202698	0.739998
18	0.985304	0.45593	0.854995	0.87599	0.775589	0.627693	0.672705	0.602002	0.586091	0.932696	0.641899	0.70037	0.137259	0.207932	0.946573
19	0.650513	0.493435	0.077174	0.222826	0.181643	0.508868	0.133042	0.202549	0.399554	0.526633	0.517332	0.181284	0.515926	0.458233	0.525616
20	0.151077	0.335909	0.634422	0.205539	0.325112	0.318502	0.399464	0.051124	0.688496	0.701267	0.132593	0.927342	0.453448	0.08489	0.129423

Figure 2. Random number sample example diagram

After generating random samples (y), return to the table of task1 and use the formula x= $\frac{-1}{\lambda}$ln(1-y) to obtain random samples of the exponential distribution Exp(λ=0.2) (see Figure 3). The sample size of each group is 100, and then the pull-down formula is repeated for 1000 times. Finally, we get a 100 x 1000 exponentially distributed random sample (see Figure 4).

=-LN(1-'Task 1(Random sample)'!A2)/0.2

Figure 3. Exponential formula

L	M	N	O	P	Q	R	S	T	U
Simulate 100 random values from the Exp(lambda=0.2) distribution and find the sample mean. Repeat this 1,000 times to observe the distribution of the sample mean									
1	2	3	4	5	6	7	8	9	10
0.715031791	2.391333427	4.802711184	4.916937661	6.175468374	5.592233309	16.96651484	3.288999164	2.132989862	2.171627874
1.167617784	1.728061819	2.032963197	4.391428878	1.95674614	0.91622859	2.400025992	2.622149063	0.455662101	10.95890592
0.398331282	4.06507906	2.906479595	14.4174761	7.830474258	0.167625962	8.797467603	0.767932505	12.03616882	2.219416617
7.694860411	0.075085648	12.11144985	0.174744424	0.630180647	1.832028286	11.91640682	2.500598834	9.844086992	5.798224463
11.78363544	2.42085294	10.71720931	8.445289505	4.114549667	3.565877558	2.145147027	6.618789748	3.912011205	4.923338503
5.420855151	9.201711103	8.193454754	4.976460464	7.405173698	7.469371691	0.958254955	1.0904001	16.43898109	2.556140202
13.43524099	0.999541643	3.61586308	0.293686255	1.429730299	2.606005661	7.640801014	1.2059171	1.21334543	18.62426574
4.162107628	0.372809779	9.347941114	0.734739072	2.618108324	4.959093525	0.970768628	1.538384924	1.92873654	2.132760766
10.62120693	3.529092249	1.478533893	10.49397226	10.4021236	1.030134703	14.10153094	4.163482921	6.194531017	2.92416132
1.249117452	4.388550426	13.43743796	5.903761831	8.056009556	1.994932302	7.427582383	1.299283115	3.120110762	4.112506955
0.835164581	4.73787533	0.876512272	9.506579575	8.702806223	0.074174911	5.273585233	2.05998377	3.28178701	7.120161942
1.917093739	0.210021659	0.370393609	10.25114585	8.954547924	8.371349306	3.676325719	1.952545733	1.429332231	11.54625571
4.26666993	14.77479921	6.731302715	6.276761519	10.27093699	2.217785209	12.98599244	0.324068569	21.79879917	0.845779411
1.019671705	8.084553775	14.52556755	4.520488461	2.205682811	6.454629602	0.367495745	19.79240993	0.088462217	1.839368099
13.62323666	10.58008887	18.03421881	3.81197457	0.104469862	6.899020199	7.986092465	12.75452021	10.47085143	0.128961439
18.24238131	2.18526833	2.450303651	1.790566999	6.825865634	5.030561848	1.868183679	4.391788801	3.537883936	0.521950795
21.10104328	3.043390098	9.654923523	10.43697233	7.471370396	4.940179804	5.584460001	4.606537839	4.410540594	13.49267842
5.256440435	3.400514772	0.401571161	1.260457131	1.002281884	3.555209331	0.713824222	1.131671938	2.550408788	3.739423556
0.818932151	2.046680407	5.031379887	1.150459518	1.966043791	1.917313164	2.549661696	0.262384138	5.831716267	6.041022449

Figure 4. Example of random samples from exponential distribution Exp(λ=0.2)

2. Sample mean calculation

We simulated 1,000 sample means, each of which was calculated by "=SUM(...)/100" based on 100 random samples drawn from each row of the exponential distribution Exp (0.2). After obtaining the 1,000 sample means, the average ("=AVERAGE(...)"), standard deviation ("=STDEV.S(...)"), and extreme values (maximum "=MAX(...)" and minimum "=MIN(...)") of the sample means were calculated using Excel formulas (see Figure 5 & 6). Since the theoretical mean of the index is μ=1/λ=5, the theoretical standard error is SE=μ/$\sqrt{n}$=0.5. The simulation results show that the actual mean and standard deviation are 4.80 and 0.43, respectively, which are very close to the theoretical values. This shows that the sample mean distribution is relatively concentrated, very close to the theoretical value, and the simulation effect is good.

Set
4.36
4.00
4.25
5.58
4.95
4.44
5.27
4.61
5.35
4.82
4.97

Minimum	3.53
Maximum	6.34
Average	4.80
SD	0.43

Figure 5 & 6. Sample mean example plot & Average, SD, and Extreme Values of the sample mean

3. Central Limit Theorem Verification

First, draw a histogram of the 1000 sample means. Select all sample means and use "Charts" > "Histogram" to generate a chart. From the histogram, we can observe that its distribution is close to a symmetrical bell-shaped curve, with its peak roughly in the middle and a nearly normal distribution (see Figure 7). When the sample size is 1000, its mean distribution shows a normal trend, which conforms to the central limit theorem.

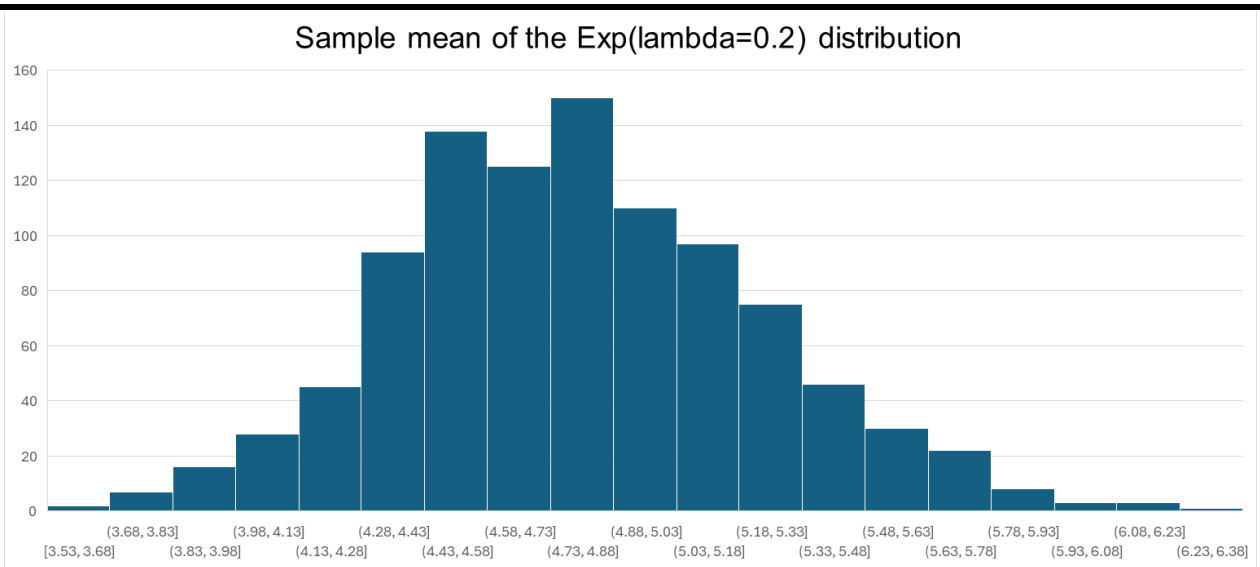


Figure 7. Histogram of sample mean of Exp(lambda=0.2)

Second, for accuracy, we also use The Quantile-Quantile (Q-Q) plot to further verify whether it conforms to the central limit theorem. Use "=SORT(...)" to re-sort the 1000 sample means in ascending order and set them as x values, then use “NORM.S.INV((ROW(...)-0.5)/1000)” to calculate the theoretical normal quantile and set it as the y value. Select these two columns of data to draw a scatter plot, use "Charts" > "Scatter" to generate a chart, and add a trend line, formula and R^2 value (see Figure 8). The Q-Q plot shows that the points fall roughly near the trend line, indicating that the sample mean distribution is approximately normal, which is consistent with the conclusion of the central limit theorem.

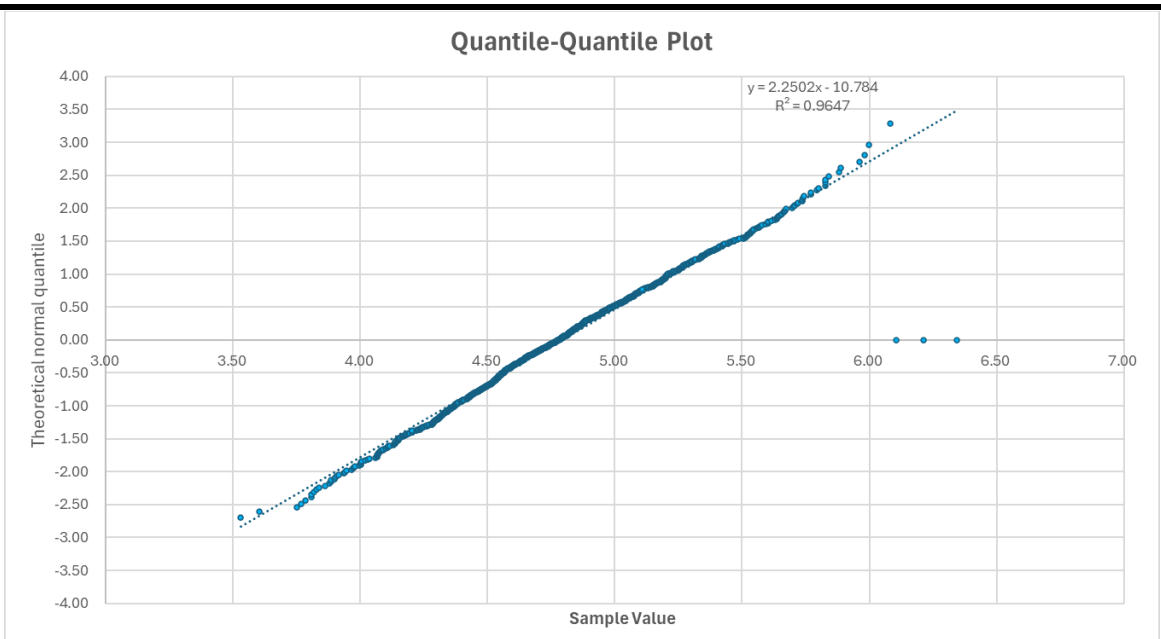


Figure 8. Quantile-Quantile plot of the sample mean

In addition, we also used the Jarque-Bera normality test to verify the central limit theorem. First, use Excel to calculate the sample mean skewness ("=SKEW(...)"), excess kurtosis ("=KURT(...)"), test statistic ($JB=\frac{n}{6}(S^2+\frac{(K-3)^2}{4})$), p-value ("=1-CHISQ.DIST(JB, 2, 1)") and critical value ("=CHISQ.INV(1-a, 2)"), and then set the hypothesis. We set the null hypothesis (H_0) that the data follows a normal distribution, and the alternative hypothesis (H_1) that the data does not follow a normal distribution. According to Figure 9, the test statistic JB = 9.6432 exceeds the critical value of 5.9915 at a significance level of 5%, and the p-value is 0.0081. This leads us to reject the null hypothesis of normality. Although the Jarque-Bera normality test result rejects the normality hypothesis (with a very small p-value) due to its extreme sensitivity to slight skewness (0.2374) and kurtosis (0.0773), the test is very sensitive to slight deviations given the large sample size. Overall, the sample means showed characteristics of a normal distribution.

Statistical test for normality (Jarque-Bera Test)				
Sample skewness, s	0.2374		Test statistic, JB	9.6432
Sample excess kurtosis, k-3	0.0773		$P(\chi^2 > JB)$	0.0081
Sample size, n	1000		χ^2 critical value	5.9915
Level of signficance, α	0.05			

Figure 9. Statistical test for normality (Jarque-Bera Test)

4.Conclusion

This task successfully simulates 1,000 groups of sample means generated from the exponential distribution Exp(λ=0.2) and shows their distribution characteristics. The results show that its distribution is close to normal, which conforms to the central limit theorem.

The histogram shows a symmetrical bell-shaped distribution, and the Q-Q plot shows that the mean distribution points basically fall near the trend line, strongly supporting approximate normality. Although the Jarque-Bera test formally rejects the normality hypothesis, this result is mainly limited by the sensitivity of the test itself to the skewness and kurtosis of large samples, but it does not affect the overall judgment.

In summary, combining graphics and descriptive statistical analysis, we can reasonably judge that the sample mean distribution is close to normal, verifying the central limit theorem: even if the original data obeys a skewed distribution, as long as the sample size is large enough, its sample mean will tend to a normal distribution.

The summary

This task simulates 1000 groups of sample means from the exponential distribution Exp(λ=0.2), plots its distribution and performs a normality test, effectively verifying the central limit theorem. The approximate normality is verified by using histograms, quantile-quantile (Q-Q) plots, and Jarque-Bera

normality tests. In the end, the sample mean is close to the theoretical value, which is consistent with the central limit theorem.

Task 2

Task 2A

To simulate the daily operation of a small local bank, we simulated the customer arrival time and service time for the remaining four working days based on the structure and assumptions established in the provided example on the first day. The business hours of this bank are from 9:00 a.m. to 4:00 p.m., totalling 420 minutes each day. During this period, a staff member will provide services to customers in the order of their arrival.

Suppose the customer arrival interval follows a "truncated" exponential distribution with a rate of $\lambda = 0.1$. The service time follows a separate “truncated” exponential distribution, $\lambda = 0.125$. This simulation first generates the arrival interval and accumulates the sum to determine the customer's arrival time, which continues until the arrival time is 420 minutes. The part exceeding 420 minutes is excluded. For each customer, the service start time is calculated using the maximum value of the arrival time and the service end time of the previous customer. And we define the waiting time as the difference between the start time of the service and the arrival time. If it is a negative number, it defaults to zero. Similarly, define the idle time of employees as the gap between the arrival time of a new customer and the end time of the previous customer service. If it is a negative number, it is also defaulted to zero. Finally, we calculate the service end time by adding the service duration to the service start time.

This simulation process ensures that each customer can receive service immediately when employees are idle or queue up when they are busy. And whenever there is no one in the bank, customers can accurately obtain the idle time of the employees. The structure and formula used in the following four working days remain consistent with those on the first day to ensure the consistency of the entire dataset. After the daily simulation was completed, we excluded all customers who arrived after 420 minutes because this was not included in the bank's business hours.

Our simulation results include detailed records of arrival time, service start and end time, waiting time and employee idle time. These datasets were subsequently produced as the shared data table "Task 2

(Data)" to support the subsequent analysis in tasks 2B to 2E.

To illustrate the simulation process and verify the validity of our simulation, the first 20 rows of the Day 2 simulation are shown in Figure 10 below.

Day 2	Time (0-420 min)	Arrival	Cumulative Arrival	Service	Completion	Arrive - Completion	Waiting	Idle	U(0.25,0.75)	U(0.25,0.75)
			0.00		0.00					
		7.03	7.03	9.02	16.05	7.03	0.00	7.03	0.50	0.68
		9.34	16.37	7.21	23.58	0.32	0.00	0.32	0.61	0.59
		7.45	23.82	10.28	34.09	0.24	0.00	0.24	0.53	0.72
		4.46	28.28	4.21	38.30	-5.82	5.82	0.00	0.36	0.41
		13.79	42.06	9.38	51.45	3.76	0.00	3.76	0.75	0.69
		9.29	51.35	6.12	57.57	-0.09	0.09	0.00	0.61	0.53
		3.55	54.91	5.32	62.89	-2.67	2.67	0.00	0.30	0.49
		8.87	63.78	3.20	66.98	0.88	0.00	0.88	0.59	0.33
		3.16	66.94	5.35	72.33	-0.05	0.05	0.00	0.27	0.49
		12.15	79.08	5.46	84.54	6.75	0.00	6.75	0.70	0.49
		8.25	87.33	5.93	93.26	2.79	0.00	2.79	0.56	0.52
		11.16	98.49	6.23	104.72	5.23	0.00	5.23	0.67	0.54
		5.89	104.38	4.09	108.81	-0.34	0.34	0.00	0.45	0.40
		7.04	111.43	4.74	116.17	2.61	0.00	2.61	0.51	0.45
		2.93	114.35	3.98	120.15	-1.81	1.81	0.00	0.25	0.39
		4.21	118.57	5.11	125.26	-1.58	1.58	0.00	0.34	0.47
		2.97	121.54	3.59	128.85	-3.72	3.72	0.00	0.26	0.36
		7.40	128.94	8.69	137.64	0.09	0.00	0.09	0.52	0.66
		10.86	139.80	4.00	143.80	2.17	0.00	2.17	0.66	0.39
		9.32	149.12	7.56	156.68	5.31	0.00	5.31	0.61	0.61

Figure 10. First 20 rows of simulated customer arrival and service data for Day 2.

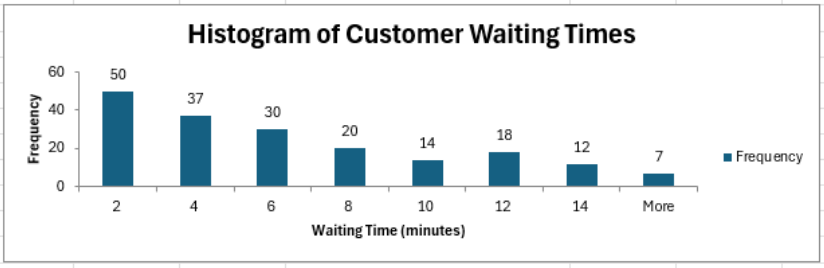
Task 2B

Customer Waiting Time

The analysis of customer waiting times reveals that the majority of customers experience short waiting periods, predominantly between 2 and 6 minutes. This is illustrated by the histogram, where the highest frequencies are within the 2-minute (n=50) and 4-minute (n=37) bins. The distribution shows a clear right-skew, as confirmed by the skewness value of 1.1961. The mean waiting time is 6.09 minutes, which is higher than the median of 4.92 minutes, indicating the presence of a few long waits (up to 26.48 minutes). The standard deviation of 5.24 minutes suggests moderate variability among the waiting periods.

Frequency Distribution of Waiting Times

Bin	Frequency
2	50
4	37
6	30
8	20
10	14
12	18
14	12
More	7



Waiting Times	
Mean	6.0901
Standard Error	0.3749
Median	4.9168
Standard Deviatior	5.2355
Sample Variance	27.4104
Kurtosis	1.4001
Skewness	1.1961
Range	26.4430
Minimum	0.0418
Maximum	26.4848
Sum	1187.5744
Count	195

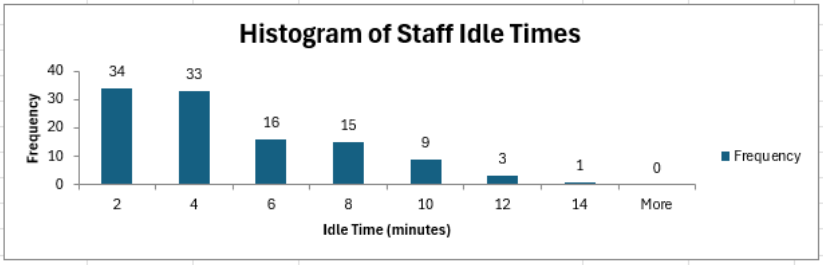
The customer waiting times are primarily concentrated between 2 to 6 minutes, as shown in the histogram. The distribution is positively skewed, indicating that while most customers wait for a short period, a few experience longer delays. This is supported by the descriptive statistics, where the mean (6.09 minutes) is higher than the median (4.92 minutes), and the skewness value of 1.20 further confirms the right-skewed nature. The standard deviation of 5.24 suggests a moderate variability in waiting times. The presence of a few long waits (up to 26.5 minutes) may warrant further investigation to improve service efficiency.

Figure 11. Histogram and Descriptive Statistics of Customer Waiting Times

Staff Idle Time

Staff idle times exhibit a similar pattern, with most idle durations between 2 and 6 minutes. However, the distribution is less skewed, with a skewness of 0.7183 and a mean of 3.90 minutes, slightly higher than the median of 3.31 minutes. The maximum idle time is only 12.09 minutes, compared to the longer tail seen in waiting times. Furthermore, the standard deviation of 2.85 minutes indicates that idle times are more tightly clustered around the mean than customer waiting times. The excess kurtosis of -0.2399 suggests a flatter shape than a normal distribution.

Frequency Distribution of Idle Times	
Bin	Frequency
2	34
4	33
6	16
8	15
10	9
12	3
14	1
More	0



Idle Times	
Mean	3.8987
Standard Error	0.2704
Median	3.3085
Standard Deviatior	2.8490
Sample Variance	8.1171
Kurtosis	-0.2399
Skewness	0.7183
Range	12.0714
Minimum	0.0149
Maximum	12.0862
Sum	432.7611
Count	111

The histogram of staff idle times displays a similar but less extreme pattern. Most idle times fall within 2–6 minutes, with frequencies gradually decreasing as time increases. The mean idle time is 3.90 minutes, slightly higher than the median of 3.31 minutes, indicating slight positive skewness (skewness = 0.72). The distribution is relatively tighter than the customer waiting times, with a smaller standard deviation (2.85) and a lower maximum value (12.09 minutes). The negative kurtosis (-0.24) suggests a flatter distribution, indicating less clustering around the mean compared to a normal distribution.

Figure 12. Histogram and Descriptive Statistics of Staff Idle Times

Comparison & Conclusion:

Both customer waiting times and staff idle times are right-skewed, meaning that although most values are small, some extreme high values exist. However, waiting times show greater skewness and higher variability, indicating that customer experiences are less consistent. In contrast, staff idle times are more stable and predictable. This suggests that while staff scheduling may be efficient, customer flow experiences could benefit from further optimization to reduce excessively long waiting periods.

Task 2C

Task 2C	
μ	5
t_cv	1.9723
Se	0.3749
B	0.7394
95% t-confidence interval for μ	
LCL	5.3507
UCL	6.8296
Hypothesis Test (Right-tailed):	$H_0: \mu = 5$
	$H_1: \mu > 5$
t	2.9076
P-value	0.0020
Decision:	
Since the p-value is less than the 0.05 significance level, we reject the null hypothesis.	
Conclusion:	
At the 5% level of significance, there is strong evidence that the mean customer waiting time is significantly greater than 5 minutes.	

Figure 13. Hypothesis test for waiting time

To improve the customer service efficiency of the bank, we use statistical inference and analysis in the non-zero waiting time data of 195 customers. Aiming to decide whether the average waiting time of customers is higher than 5 minutes. This research collected 195 non-zero waiting time values.

We built the 95% confidence interval [5.3507, 6.8296] of the average customer waiting time. The real customer waiting time may exceed the target standard because the interval does not include the 5-minute lag in the null hypothesis. We have 95% confidence that the average waiting time of customers falls in this range.

1. This research question can be formulated as a right-tailed hypothesis test with the population mean μ:

Null Hypothesis (H_0): μ=5 (The customer average waiting time is equal to 5 minutes)

Alternative Hypothesis (H_1): μ>5(The average customer waiting time is larger than 5 minutes)

2. Because the population standard deviation is unknown and the sample size is large enough (n = 195), the Central Limit Theorem shows the sampling distribution of the sample mean is approximately

normal. Additionally, while the simulated data reflects a typical operating day, real-life variability such as special events or staff shortages are not captured, limiting the generalisability of this analysis. Therefore, this test selected the t-test. We adopted the t-distribution with degrees of freedom of n-1 = 194 for inference.

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1} \text{ under } H_0$$

So, the t-statistic is calculated as follows: (6.0901-5）/0.3749=2.9076

3. The commonly used significance levels were selected for this study: α = 0.05, a 5% allowable error probability.

4. Utilize Excel T.INV(0.95, 194).The critical value:t 0.05,194=1.9723

Decision rule: Reject H_0 if the calculated t-statistic is larger than 1.9723.

5.t_statistic=2.9076

p_value = 0.0020

The t-statistic is higher than the critical value 1.9723. The p-value is smaller than the significance level of 0.05. There is sufficient evidence to reject the null hypothesis.

The 95% confidence interval for the mean is [5.3507, 6.8296], which is higher than the 5-minute standard. This gives more support.

Conclusion: The average waiting time of customers was higher than 5 minutes at the 5% significance level. This means that the current bank service distribution cannot satisfy the consumers' immediate service demands, especially during peak customer flow periods. The bank should improve staff scheduling or add an appointment system to strengthen customer satisfaction and operational efficiency.

Task 2D

Task 2D	
μ	5
t_cv	1.9818
Se	0.2704
B	0.5359
95% t-confidence interval for μ	
LCL	3.3628
UCL	4.4347
Hypothesis Test (left-tailed):	$H_0: \mu = 5$
	$H_1: \mu < 5$
t	-4.0724
P-value	0.0000
Decision:	
The t-value is significantly less than the critical value, and the p-value is below 0.05, we reject the null hypothesis.	
Conclusion:	
Therefore, we reject the null hypothesis and conclude that the mean staff idle time is significantly less than 5 minutes.	

Figure 14. Hypothesis test for Idle time

This study uses statistical inference and analysis of the employer's non-zero idle time data in order to evaluate the efficiency of banks in the distribution of customer service resources. The research goal is to decide whether the average idle time of employees is lower than 5 minutes. We can know if the bank has an overwork problem. 111 valid non-zero idle time records in this simulation.

We constructed the 95% confidence interval [3.3628, 4.4347] of the average idle time of employees. This range is lower than the 5-minute standard set in the null hypothesis, indicating that the actual idle time of employees is likely to be lower than the target value. We have 95% confidence that the average idle time of employees falls in this range.

1. This research question is formulated as a left-tailed hypothesis test concerning the population mean (μ)

Null Hypothesis (H_0): $\mu = 5$ (The average staff idle time is 5 minutes)

Alternative Hypothesis (H_1): $\mu < 5$ (The average staff idle time is less than 5 minutes)

To test whether employees are under high work pressure because of insufficient idle time.

2. Since the population standard deviation is unknown and the sample size is large enough (n = 111), based on the central limit theorem, it can be considered that the sample mean approximately follows a normal distribution. Additionally, our simulation assumes consistent staff behaviour and arrival patterns across days, which may not reflect actual variability in real bank operations. Therefore, the t-test is adopted. Based on the t-distribution, the degree of freedom is df = 110 for inference.

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1} \text{ under } H_0$$

So, the t-statistic is calculated as follows: (3.8987-5）/0.2704=-4.0724

3. This study adopts the commonly used significance levels: α = 0.05

4.Utilize the Excel T.INV(0.05, 110). The critical value: $t_{0.05,110}$ = -1.9818
Decision rule: Reject H_0 if the calculated t-statistic is lower than -1.9818.

5.t_statistic=-4.0724
p_value = 0.0000

Because the t-statistic is less than the critical value of -1.9818. The p-value is lower than the 0.05 significance level. There is sufficient evidence to reject the null hypothesis.

In addition, the 95% confidence interval for the mean staff idle time is [3.3628, 4.4347], which is fully below the 5-minute standard. This gives more support to the result that the average idle time is less than 5 minutes.

Conclusion: At the significance level of 5%, the average idle time of employees was significantly less than 5 minutes. This shows that bank employees are busy most of the time, with almost no extra time to relax. This may cause a high workload problem. It is suggested that the bank management review the current shift scheduling system, appropriately increase the distribution of human resources, or

introduce business process automation tools to reduce employee pressure, improve service quality, and prevent potential efficiency decline or employee burnout.

Task 2E

Hypotheses:

Null Hypothesis (H_0): $\mu_1 = \mu_2$ (Mean Waiting Time = Mean Idle Time)

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (Mean Waiting Time ≠ Mean Idle Time)

Results of Two-Tailed t-Test:

According to known information

Waiting Times		Idle Times	
Mean	6.0901	Mean	3.8987
Standard Error	0.3749	Standard Error	0.2704
Median	4.9168	Median	3.3085
Standard Deviation	5.2355	Standard Deviation	2.8490
Sample Variance	27.4104	Sample Variance	8.1171
Kurtosis	1.4001	Kurtosis	-0.2399
Skewness	1.1961	Skewness	0.7183
Range	26.4430	Range	12.0714
Minimum	0.0418	Minimum	0.0149
Maximum	26.4848	Maximum	12.0862
Sum	1187.5744	Sum	432.7611
Count	195	Count	111

Figure 15. Descriptive Statistics Summary of Customer Waiting Times and Staff Idle Times

t statistic = $t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ =4.7405

Degrees of freedom = $df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$ = 303

p-value (two-tailed) = 0.0000033

Critical t-value = ±1.9678 (at α = 0.05 significance level)

Decision：

Since the calculated t-value (4.74) exceeds the critical value (±1.9678), and the p-value is effectively zero, we reject the null hypothesis. There is strong statistical evidence that the mean customer waiting time is significantly different from the mean staff idle time.

Conclusion：

There is strong statistical evidence that the mean customer waiting time is significantly different from the mean staff idle time. The longer and more variable waiting times suggest inefficiencies in customer service operations, despite relatively stable idle times among staff. This finding underscores the need for resource reallocation or scheduling adjustments to better align staff activity with customer traffic, improving overall service efficiency.

The summary

Analysis of Tasks 2A–2E shows that customer waiting times are more variable and significantly longer than staff idle times. The right-skewed distributions and t-test results confirm inefficiencies in service flow. These findings highlight the need to optimize staff allocation to reduce long customer waits without increasing idle time.